

Prateek Verma

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Seeking opportunities where I can leverage my industrial and academic experience to drive differentiation and innovation. I am a machine learning scientist with cross-functional expertise in chemicals, materials, health, and environment. My key strengths lie in visionary and complex problem solving, interdisciplinary research, leadership, and research advising.

Experience

Manager, Data Science Core, University of Arkansas, 2023–present

- Evaluation, fine-tuning, and prompt engineering of large (vision) language models (LLMs) using multimodal data
 - Leading machine learning and data science support for 50+ researchers focusing on multimodal and multi-omic medical, chemical and materials research
 - Managing high-performance computing and data-storage servers
 - Example projects: AI for microscopy, AI for materials science, cell movement and division tracking, cell viability prediction, AI for women's health, multi-omics and ANOVA, animal pose tracking and behavior classification, data mining and analysis from PubMed
- **Postdoctoral Fellow, University of Arkansas, 2021–2023**
- Built an end-to-end CNN ML pipeline for scientific images
 - Built graph and generative algorithms for molecular discovery and finding functional groups on molecules and macromolecules
 - Applications: predicting onset of diseases, sensors for bacteria and viruses, drug and molecular design, medical diagnosis
- **Postdoctoral Fellow, Georgia Institute of Technology, 2018–2021**
- Developed multivariable deep neural network regression to split, interpolate, and predict total signal into constituents
 - Developed CNN and regression algorithms for noise detection in signals
 - ML applications: Extract pollution composition (expensive measurement) from total PM2.5 (inexpensive) data; noise detection in light scattering data
 - Fabricated metamaterial composites using tensegrity/auxetic approaches
 - Executive Director for OPALL (Open Polymer Active Learning Laboratory)
- **Senior Coating Chemist, Kimoto Tech, 2016–2018**
- Team leader for 5 R&D chemists
 - Led scale-up and production of 15+ lab-to-market products
 - Development of flexible & protective coatings, conductive coatings, and pressure sensitive adhesives
- **PhD candidate, Georgia Institute of Technology, 2011–2015**
- Synthesized intrinsically auxetic liquid crystal elastomers
 - Developed protocols for accurately measuring Poisson's ratio
 - Developed a new method to induce auxetic properties in nonwovens
 - Modeled processing-structure-property relations for auxetic behavior in fiber networks
- **Research Assistant (Master's), University of Akron, 2011**
- Developed a buckling-based metrology to determine strength of thin films
 - Prepared and tested strength of polymer-blend films used in tissue engineering
- **Summer Intern, University of Massachusetts Amherst, 2010**
- Synthesized uniformly sized silica nanoparticles for composite applications
 - Created super-hydrophobic surfaces using silanes; synthesized cross-linked silicones
- **Summer Intern, University of Minnesota, 2009**
- Synthesized epoxy networks to study flow through kidney membranes
 - Synthesized and characterized collagen gels for tissue engineering

Education

PhD, Georgia Institute of Technology, 2011–2015, Materials Science and Engineering

BS and MS, Indian Institute of Technology Roorkee, 2006–2011, Polymer Science and Technology

Skills

Machine Learning: convolutional neural networks, graph neural networks, image preprocessing, k-Means clustering, linear regression, logistic regression, ML pipelines, chemical informatics, large language models, RESNET, support vector machines,

vision language models

Computer languages: C/C++, MATLAB, PHP, Python, SQL

Computational: MATLAB, AWS, Django, LAMMPS, Matplotlib, MySQL, Numpy, Pandas, RDKit, Tensorflow & PyTorch, Scikit, BioPython

Data Infrastructure: high performance computing, GPU computations, data pipelines, data mining, ML workflows, cloud, dashboards

Chemistry: FR polymerization, LCE synthesis, polyurethane synthesis, silanes & silicones, thermal & UV curing

Materials: auxetics, biopolymers, characterization, liquid crystals, metamaterials, nanotechnology, polymer processing, structure-property relationships, thermal analysis, viscoelasticity

Industry: adhesives, process development, chemical mixing, chemical formulations, protective coatings, scale-up operations, thermal & UV curing

Interpersonal: illustration, leadership, mentoring, research advising, teaching, team building

Lab Techniques: AFM, DSC, TGA, DMA, environmental testing, FTIR, isothermal titration calorimetry, mechanical testing, micro-CT, electron microscopy, viscometry

Select Publications

Please visit [Google Scholar](#) or my [website](#) for a complete list of my 20+ publications, 20+ conference presentations, and 300+ citations.

1. P Verma, C Kuebler, MH Van, et al.; Machine learning powered tracking of cells, midlines, cell divisions and their angles during embryo development; eLife; **2025**.
2. P Verma, MH Van, C McEnaney, et al.; A machine learning framework for automated computational ethology using markerless pose estimation; NeurIPS Workshop and IEEE BigData – 8th Special Session on HealthCare Data; **2025**.
3. A Vuong, MH Van, P Verma, et al.; Fine-tuning vision-language models for multimodal polymer property prediction; IEEE BigData – 8th Special Session on HealthCare Data; **2025**.
4. A Carey, VM Shashidhar, ..., P Verma, et al.; Development of an Integrated Multi-Omics Data Platform for Women's Precision Health; IEEE BigData; **2025**.
5. MH Van, P Verma, X Wu; A Survey of AI for Materials Science: Foundation Models, LLM Agents, Datasets, and Tools; arXiv; **2025**.
6. P Verma, MH Van, X Wu; Beyond human vision: The role of large vision language models in microscope image analysis; IEEE BigData; **2025**.
7. P Verma, E Adeogun, ES Greene, et al.; Machine-learning based sensing of heat-stress in organisms; RSC Sensors Diagnostics; **2024**.
8. MH Van, P Verma, X Wu; On large visual language models for medical imaging analysis: an empirical study; IEEE/ACM CHASE; **2024**.
9. D Ansari, P Verma, T Ansari; Promise of machine learning techniques towards retrieving aerosol chemical composition from temporal variations of total PM mass concentrations; Journal of Research in Atmospheric Science; **2023**.
10. H Sun, X Fang, ..., P Verma, et al.; An ultra-sensitive and stretchable strain sensor based on micro-crack structure for motion monitoring; Nature Micro Nano (IF=9.9); **2022**.

Honors and awards

- NISBRE Merit Awardee, U Arkansas (2024–present)
- Lead scientist on \$300,000 award, U Arkansas (2021–2023)
- MSE 5 year mentorship award, Georgia Tech (2021–present)

- Hightower Fellow, OPALL, Georgia Tech (2019–2021)
- Chairman, Tech. Conference, Kimoto, Kimoto Tech (2017–present)
- Second prize, auxetic conference, Georgia Tech (2014–present)

Leadership experience

- DEI council representative for MSE staff, Georgia Tech (2019–2021)
- Co-launched Postdoc Chats, Georgia Tech (2019–2023)
- Advisor/mentor for OPALL members, Georgia Tech (2019–2022)
- Team leader, Kimoto Tech (2016–2018)
- Co-manager for thermal analysis lab, Georgia Tech (2013–2015)
- Student President (elected), IIT Roorkee (2008–2009)
- Founder, web dev team, IIT Roorkee (2007–2010)